Algorithm no-**2.1**

Algorithm Name: **Traversing in Array from Lower bound to Upper Bound**

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define newl cout<<endl
int main()
{
    vector<int>A={7,9,3,5,4,9,10,12,15,16,18,25,22,29};
    for(int i:A)
    {
        cout<<i<<" ";
    }
    newl;
    int a,b;
    cin>>a>>b;
    for(int i=a-1;i<b;i++)
    {
        cout<<A[i]<< " ";
    }
}
```

Output:

```
7 9 3 5 4 9 10 12 15 16 18 25 22 29
4 10
5 4 9 10 12 15 16
Process returned 0 (0x0)   execution time : 1.783 s
Press any key to continue.
```

Algorithm no-**2.2**

Algorithm Name: **Inserting in an Array in a specific position**

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define newl cout<<endl
int main()
{
    vector<int>A={7,9,3,5,4,9,10,12,15,16,18,25,22,29};
    for(int i=0;i<14;i++)
    {
        cout<<A[i]<<" ";
    }
    newl;
    int n=A.size();
    int a,b;
    cin>>a>>b;
    for(int i=n;i>=a;i--)
    {
        swap(A[i],A[i-1]);
    }
    A[a-1]=b;
```

```
    for(int i=0;i<=n;i++)
    {
        cout<<A[i]<<" ";
    }
}
```

Output:

```
7 9 3 5 4 9 10 12 15 16 18 25 22 29
5 8
7 9 3 5 8 4 9 10 12 15 16 18 25 22 29
Process returned 0 (0x0)   execution time : 1.151 s
Press any key to continue.
```

Algorithm no-**2.3**

Algorithm Name: **Inserting in an Array in a specific position**

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define newl cout<<endl
int main()
{
    vector<int>A={7,9,3,5,4,9,10,12,15,16,18,25,22,29};
    for(int i=0;i<14;i++)
    {
        cout<<A[i]<<" ";
    }
    newl;
    int n=A.size();
    int a;
    cin>>a;
    for(int i=a-1;i<n;i++)
    {
        swap(A[i],A[i+1]);
    }
    for(int i=0;i<A.size()-1;i++)
    {
        cout<<A[i]<<" ";
    }
}
```

Output:

```
7 9 3 5 4 9 10 12 15 16 18 25 22 29
9
7 9 3 5 4 9 10 12 16 18 25 22 29
Process returned 0 (0x0)   execution time : 0.984 s
Press any key to continue.
```

Algorithm no-**2.3**

Algorithm Name: **Ascending sorting using Bubble Sort Algorithm**

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define newl cout<<endl
int main()
{
   vector<int>A={7,9,3,5,4,9,10,12,15,16,18,25,22,29,0};
   int n=A.size();
   for(int i=0;i<n;i++)
   {
      cout<<A[i]<<" ";
   }
   newl;
   while(1)
   {
      int a=0;
      for(int i=0;i<n-1;i++)
      {
         if(A[i]>A[i+1])
         {
            swap
            (A[i],A[i+1]);
         }
         else
         {
            a++;
         }
      }
      if(a==n-1)
      {
         break;
      }
   }
   for(int i:A)
   {
      cout<<i<<" ";
   }
   newl;
}
```

Output:

```
7 9 3 5 4 9 10 12 15 16 18 25 22 29 0
0 3 4 5 7 9 9 10 12 15 16 18 22 25 29

Process returned 0 (0x0)   execution time : 0.040 s
Press any key to continue.
```

Problem No- **Round 890 (DIV 2 A)**

Problem Name: **Tales of a sort**

Problem Link: **https://codeforces.com/contest/1856/problem/A**

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define _easy_ ios_base::sync_with_stdio(0);cin.tie(0);cout.tie(0);
#define ll long long int
#define A_sort(x) sort(x.begin(),x.end())
#define newl cout<<endl
int main()
{
    int t;
    cin>>t;
    while(t--)
    {
        int n,a=-100,x=0;
        cin>>n;
        pair<int,int>pr;
        vector<int>A(n),B;
        for(int i=0; i<n; i++)
        {
            cin>>A[i];
            if(a<=A[i])
            {
                pr.first=A[i];
                a=A[i];
                pr.second=i;
            }
        }
        B=A;
        A_sort(B);
        if(B==A)
        {
            cout<<0;
            newl;
            continue;
        }
        else if(pr.second>=0&&pr.second<n-1)
        {
            cout<<a;
        }
        else
        {
            for(int i=n-1;i>=0;i--)
            {
                if(A[i]==B[i])
                {
                    x++;
                }
                else
                {
                    break;
                }
            }
            cout<<*max_element(A.begin(),A.begin()+(n-x));
        }
        newl;
    }
}
```

Output:

217310859	00:47:07	Arsenic33	A - Tales of a Sort	GNU C++20 (64)	Accepted	15 ms	16 KB

Problem No- Introductory Problem

Problem Name: **Missing Number**

Problem Link: **https://cses.fi/problemset/task/1083**

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define newl cout<<endl
int main()
{
   ll n;cin>>n;
   set<int>s;
   for(int i=1;i<=n;i++)
   {
      s.insert(i);
   }
   for(int i=1;i<=n-1;i++)
   {
      int a;cin>>a;
      s.erase(a);
   }
   for(auto i:s)
   {
      cout<<i;
   }
}
```

Output:

Task:	Missing Number
Sender:	Arsenic
Submission time:	2023-03-31 15:01:43 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Problem No- Introductory Problem

Problem Name: **Increasing Array**

Problem Link: **https://cses.fi/problemset/task/1094**

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define newl cout<<endl
int main()
{
   int n;
   cin>>n;
   vector<int>A(n);
   for(int i=0;i<n;i++)
   {
      cin>>A[i];
   }
   ll x=0;
   for(int i=1;i<n;i++)
   {
      if(A[i]<A[i-1])
      {
         x+=(A[i-1]-A[i]);
         A[i]=A[i-1];
      }
   }
   cout<<x;
}
```
Output:

Task:	Increasing Array
Sender:	Arsenic
Submission time:	2023-08-05 09:52:05 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED